

**LEARNSYNC: AN OFFLINE LAN-BASED LEARNING MANAGEMENT
SYSTEM WITH AI ACADEMIC ASSISTANT**

A Capstone Project
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CHAPTER I

THE PROJECT AND ITS BACKGROUND

INTRODUCTION

Education has always been a cornerstone of human progress, but in the 21st century, digital innovation has reshaped its delivery and accessibility. Cloud-based learning exemplifies this transition, enabling students and teachers to access resources anytime, anywhere, and on any device. Learners develop digital literacy skills such as online collaboration and resource management, while educators enhance their competencies through engaging the lessons. Jambhulkar (2025). Artificial intelligence has advanced education through personalized learning, adaptive feedback, and intelligent tutoring systems. However, these benefits often rely on cloud infrastructure, which disadvantages students in regions with poor internet access and deepens digital inequality. Offline AI models could help bridge this gap, but current solutions either sacrifice accuracy or require high computational power, making them unsuitable for low-resource environments. The key challenge is developing offline systems that balance efficiency and accuracy for practical, widespread use in education. Masum et.al (2025).

The research will be conducted at Palangon High School, located in Barangay Palangon Municipality of Libmanan, Province of Camarines Sur. The school is a public secondary institution under the supervision of the Department of Education (DepEd) Division of Camarines Sur with Mr. Roberto B. Malazarte HT-II/ School head. It caters to senior high school students from Palangon and neighboring barangays, serving as the primary educational institution the area.

Palangon High School provides a range of academic programs aligned with the K-12 curriculum, including core subjects and specialized strands for senior high school. For this study, the identified client Ma. Christina L. Quijano, A grade 12 teacher at Palangon High School, who plays a vital role in guiding senior high school students and supporting the integration of innovative learning approaches such as e-learning.

Palangon High School currently facing problem about low internet connectivity and weak signal in the area. Limited access to stable online platforms further restricts the use of digital tools, hinders communication, and prevents the effective integration of technology into teaching and learning. Teachers are required to manually reproduce and distribute printed modules, a process that consumes substantial time and financial resources while straining the school's limited budget. The absence of a centralized system for managing digital files forces reliance on fragmented platforms such as USB drives, email, and messaging applications. This disorganized approach complicates access to essential instructional materials, increases the likelihood of duplication or misplacement, and delays the smooth execution of academic activities. The Palangon High school also lacks timely mechanisms to identify struggling students, resulting in delayed academic interventions and widening learning gaps.

The researcher wants to improve the learning gap by introducing LearnSync, an offline LAN-based Learning Management System equipped with an AI academic assistant that directly addresses the challenges faced by Palangon High School. Unlike cloud-dependent platforms, LearnSync operates through a

local server within the school premises, allowing teachers and students to connect via LAN-enabled devices without relying on internet connectivity, thereby ensuring uninterrupted access to digital resources even in areas with weak or unstable signals. By centralizing instructional materials, teachers can upload lesson modules, quizzes, and activity sheets to the system, which students can then access smoothly. The AI academic assistant embedded in LearnSync is designed to function as a lightweight, curriculum-aligned support tool trained on school-approved modules, enabling students to search and retrieve content efficiently, receive adaptive feedback personalized to their performance, engage in interactive questioning and guided explanations that simulate a tutoring experience, and benefit from generated summaries and highlighted key concepts that reduce cognitive overload and promote comprehension. At the same time, the assistant supports teachers in the early identification of struggling learners by analyzing student activity patterns and quiz results, allowing timely interventions to close learning gaps. By integrating these features, the academic assistant not only enhances student learning but also reduces teacher workload, making instructional delivery more efficient. LearnSync minimizes reliance on printed modules, saving both time and financial resources, while promoting sustainability and efficiency in teaching. Designed to be lightweight and scalable, the system can be implemented across classrooms and extended to community-based learning hubs, empowering Palangon High School to integrate digital innovation without the barriers of internet dependency. In this way, LearnSync provides a practical, cost-effective, and curriculum-aligned solution that enhances teaching

and learning in low-resource environments. The AI academic assistant will be powered by the Mistral 7B model, ensuring efficient performance while maintaining accuracy in offline settings. Aside from that, it reportedly differs on performance compare to any Llama models on a wide range of benchmarks. Mistral 7B significantly outperforms various Llama AI models with higher parameters. Thus, choosing this model not only can be superior in code and reasoning benchmark but also has lower hardware requirements that is efficient for this project. (Mistral.ai, 2023).

General Objectives

To design and develop cloud-based Learning Management System with an AI academic assistant that enables Senior High School students to access educational materials.

Specific Objectives

1. To develop an accessible learning management system that allows students to access modules, quizzes, and learning materials.
2. Enable teachers to upload modules, create quizzes, assign activities, and record student scores.
3. To Implement a local AI academic assistant that retrieves and generates responses based solely on teacher-uploaded materials.
4. To Provide a student performance monitoring module that enables teachers to track submissions, quiz results, and academic progress.

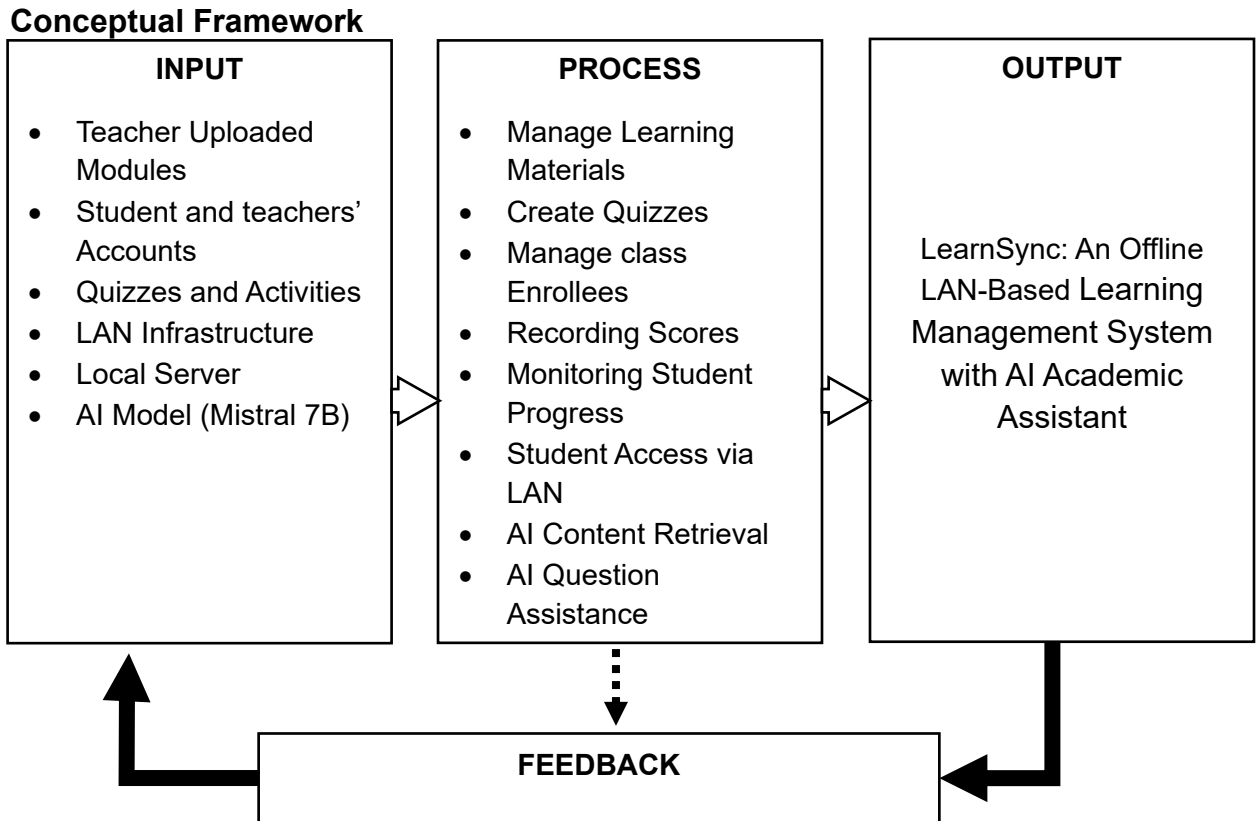


Figure 1
Learnsync: An Offline Lan-based Learning Management System with AI Academic Assistant

The Input data in LearnSync includes teacher-uploaded modules, quizzes, and activity sheets, along with student and teacher accounts connected through the school's LAN. These inputs are supported by a local server and the Mistral 7B AI model, which ensures offline functionality. In addition, the system records performance data such as quiz scores, submissions, and student activity patterns. These inputs are essential because they capture both instructional content and learner behavior, forming the foundation for adaptive feedback, performance monitoring, and curriculum-aligned support.

The Process consists of two interconnected components the Learning Management System and the AI Academic Assistant. The LMS enables teachers

to upload materials, create quizzes, assign activities, and record scores, while students access these resources through LAN-enabled devices. The system tracks submissions, quiz results, and progress, ensuring that teachers can identify struggling learners early. Meanwhile, the AI assistant retrieves and summarizes teacher-uploaded content, provides adaptive feedback, highlights key concepts, and offers guided explanations that simulate tutoring. By combining structured academic management with lightweight AI support, the process ensures that learning remains interactive, efficient, and accessible even without internet connectivity.

The Output reflects the educational and operational outcomes of LearnSync. Students benefit from uninterrupted access to centralized learning materials, enhanced comprehension through AI support, and reduced reliance on printed modules. Teachers gain efficiency by managing resources in one platform and receiving timely insights into student performance. The system produces measurable learning indicators through reports and analytics, aligning with the project's objectives of bridging learning gaps, reducing workload, and promoting sustainability. Collectively, these outputs demonstrate how LearnSync transforms the challenges of Palangon High School into a practical, inclusive, and curriculum-aligned solution for senior high school education.

The Feedback loop ensures continuous improvement by analyzing student performance data and returning insights to teachers. This allows instructional strategies and materials to be refined based on learner needs, creating a cycle of adaptation and growth. In this way, LearnSync not only supports current teaching

and learning but also evolves with each iteration, strengthening its role as a sustainable educational innovation.

Scope and Delimitation

The scope of this project is the development of a LAN-based Offline AI Learning Assistant designed. It will allow teachers to upload text-based materials such as PDFs and Word documents, create quizzes, assign activities, and record student scores. Students will access these resources and interact with a local AI assistant that generates responses strictly from the uploaded content. A performance monitoring module will also be included to help teachers track submissions, quiz results, and academic progress, with access limited to mobile and desktop devices connected to the school's Wi-Fi. For this study, the system will be implemented exclusively for Grade 12 students and will focus on the subject of Research, ensuring that the AI assistant provides curriculum-aligned support tailored to this specific academic requirement.

The delimitation of the project is that it will operate strictly offline, with no internet-based resources or external content. Access will be limited to the school's Wi-Fi range, excluding remote use. Supported materials will be restricted to text-based formats, excluding multimedia files such as videos. The AI assistant will only summarize and reframe uploaded content without generating new knowledge. The system will not include other grade levels such as Junior High and Grade 11.

Significance of the Study

The LearnSync system is expected to generate meaningful benefits to the following:

Teachers. The system helps by saving time and resources by uploading modules, creating quizzes, and monitoring student performance in one cloud-based platform.

Students. This study will have consistent access to the lesson and activities, with AI assistant that supports comprehension using only teacher upload materials.

Researcher. The study helps the researcher by enhancing technical expertise, contributing new knowledge to educational technology.

Future Researchers. The result of this study can use this project as a reference for integrating cloud-based systems and AI in education, guiding improvement and application.

Definition of terms

Learning Management System. The backbone of online education, streamlining the process of delivering and managing training programs. LMS serves as the central hub for Palangon High School's senior high students, streamlining distribution of modules, quizzes, and performance monitoring.

Local Area Network. Group of connected devices within a localized area. Allows students and teachers to access LearnSync through the school's Wi-Fi without internet.

Offline System. software program that can function without a reliable and consistent internet connection. Runs on a local server at Palangon High School, allowing students and teachers to access materials, quizzes, and AI support without internet.

Artificial Intelligence Assistant. Helpful software application that uses artificial intelligence to perform tasks. It answers student questions based only on teacher-uploaded materials.

Performance Monitoring Module. Operations practice that automatically observes key metrics of application and system to ensure services are available and reliable. Helps teachers monitor quiz results, submissions, and student progress.

NOTES

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents the review of related literature and studies collected from a range of credible references. It provides an organized discussion of existing works that are directly relevant to the topic, highlighting their contributions and insights. The review is arranged according to specific areas of concern, ensuring that each source is examined in relation to its relevance to the study.

Learning Management System

The relationship between Learning Management System (LMS) usage and academic performance among 188 Filipino senior high school students. In today's technology-driven education, institutions adopt ICTs, LMS platforms, and blended learning to improve student outcomes. However, findings based on the Pearson correlation coefficient revealed no significant relationship between LMS usage and academic performance. This highlights that while LMSs provide access and support, their effectiveness in directly influencing student achievement remains uncertain, pointing to the need for further investigation into how LMS integration impacts learning results. Tus et.al (2021). Technology has transformed education by moving beyond traditional classroom settings into interconnected, flexible learning environments. Modern learners increasingly reject rigid, stand-and-deliver approaches tied to fixed times and locations, favoring accessible, mobile, and personalized modes of participation. Advances in communication technologies and mobile devices have enabled learners to access knowledge remotely, while software vendors, open-source developers, and educational institutions have embraced systems that manage courses and engage students online. This shift

highlights the growing importance of digital platforms in meeting the evolving demands of global education. Turnbull et.al (2020). The study of Rabiman et.al (2020) highlights the weaknesses of traditional learning, where educators and students face limitations in repeating material, note-taking, and access to knowledge restricted by time and space. With students increasingly dependent on digital content, online learning offers positive opportunities for accessible, storable, and shareable teaching materials. Using the Hannafin and Peck model (needs analysis, design, development, and implementation), researchers developed and tested an LMS-based e-learning system in a Mechanical Engineering Education microteaching class. Validated by experts and tested with 15 undergraduate students, the LMS was rated “very feasible” based on usability, functions, communication, design, and content. Findings show that LMS use improves satisfaction and learning quality, demonstrating its effectiveness in modern education.

The study of Obana (2021) Learning Management System: An Essential Tool to Enhance Remote Education. The Learning Management Systems (LMS) are essential in education and corporate learning because they provide flexible, accessible, and organized platforms for delivering online and blended learning. For students, LMS enables anytime, anywhere learning, allowing them to access courses, track progress, manage assignments, and interact with professors worldwide. Teachers benefit by uploading reusable materials, creating tests, giving real-time feedback, and monitoring student performance through analytics. A unique feature, multi-tenancy, allows schools to share resources while maintaining

individual branding. In corporate settings, LMS supports employee development by tracking competencies and ensuring compliance, especially in regulated industries. Overall, LMS enhances efficiency, accessibility, and collaboration, making it a vital tool for modern education and workforce training. Rosario and Dias (2022) Indicates That the Learning Management System (LMS) is a vital tool in education as it manages, tracks, and delivers learning materials while organizing student records. Its adoption has revolutionized education by offering a comprehensive view of teaching, learning, and performance processes across students, teachers, and administrators. A systematic bibliometric literature review (LRSB) of 49 SCOPUS-indexed documents revealed the growth and innovations in LMS, identifying both opportunities and challenges in implementation. Findings emphasize that LMS designers and developers must understand the needs of different user groups to ensure tools and resources align with their expectations, ultimately improving student outcomes and learning quality.

AI Assistant

Artificial Intelligence (AI) and Knowledge Management (KM) are interconnected, with AI enabling machine learning and KM supporting knowledge sharing. A systematic review using PRISMA and PICO found that communication, trust, information systems, incentives, and organizational structure influence knowledge sharing. Integrating AI into KM enhances knowledge cycles, enterprise performance, and distribution, with leadership and organizational arrangements ensuring stability. The study, the first to examine KM with AI, calls for future research on smart assistant systems to strengthen competitiveness and deliver social benefits. Pai et.al (2021). Swetha et al. (2025) introduced Smart Assist, an

AI-based academic assistant using NLP, machine learning, and cloud computing to provide secure, personalized student support with features like authenticated access, interactive chat, cloud storage, and 24/7 assistance. Built-in analytics enhance performance and organization, while AI integration has transformed Learning Management Systems (LMSs) through personalization and distance learning. A PRISMA-2020 bibliometric analysis shows rapid growth in AI-LMS research, led by authors such as Li J and Cavus N, with strong contributions from China and India. Despite advances in e-learning and emerging themes like learning styles and ontology, gaps remain in the effectiveness and acceptance of virtual assistants, highlighting the need for future research.

Learning Assistant Management and Builder (LAMB) is an open-source framework that integrates AI assistants into Learning Management Systems (LMS) while ensuring privacy, policy alignment, and reliable information use. Its modular design and features such as prompt engineering, retrieval-augmented generation, and knowledge base creation allow educators to build custom assistants without coding. Validated through the “Macroeconomics Study Coach,” LAMB has shown potential to enhance teaching, increase student engagement, and deliver personalized learning experiences. Its scalability, security, and open-source nature further promote collaboration and innovation in AI-driven education (Forment et al., 2025). Raghavendrachar et.al (2023) Developed AI-powered Learning Management System (LMS) for web development and coding, integrating a chatbot like ChatGPT to enhance learning efficiency. The LMS features document and video modules, step-by-step project guides, career roadmaps, and resources

for problem solving and interview preparation. Its main goal is to evaluate how AI assistants accelerate skill acquisition, support app development, and improve learners' employability. The paper assesses the effectiveness of each feature, offering a holistic view of how technology can streamline learning and career readiness.

The advent of Artificial Intelligence (AI) has transformed Learning Management Systems (LMSs), enabling personalization and distance education. Using PRISMA-2020, this study applies bibliometric analysis to examine AI integration in LMSs, revealing rapid growth with a 99.42% increase, especially in 2015 and 2021. Influential contributions include authors Li J and Cavus N, the journal *Lecture Notes in Computer Science*, and countries such as China and India. Thematic clusters highlight e-learning, ontology, and ant colony optimization, while emerging terms like Learning Style and LMS suggest areas for further study. Despite progress, gaps remain in the effectiveness and acceptance of virtual assistants, underscoring the need for a future research agenda. Celerina et.al (2025).

Offline Application

The study of Laksmi et.al (2025) introduces a user-friendly educational app designed for students with limited or inconsistent internet access. The app allows lessons to be downloaded during online sessions and studied offline using government-provided tablets, addressing connectivity issues in underserved areas. Its main goal is to ensure continuous learning opportunities for students who face interruptions due to mobility or financial constraints. By bridging the digital

divide, the app provides a practical and cost-effective solution that empowers learners to access educational content anytime, fostering inclusivity and accessibility in diverse learning environments. Ahillon Jr. and Aquino (2024) study examined offline mobile applications as solutions to challenges in returning to traditional school schedules, including limited classrooms, teacher shortages, and demand for remote learning. Using a descriptive-analytic design with 2,456 students and 189 teachers in Cardona District (2023–2024), results showed that offline apps enable self-paced learning, improve accessibility, engagement, and usability, and provide alternatives for schools lacking physical modules. Findings highlight that offline mobile applications are crucial in shaping flexible, inclusive, and future-ready education.

EduVision AR, an offline augmented reality (AR) app designed to enhance interactive learning by combining physical spaces with digital experiences. Developed with Unity 3D and Vuforia SDK, the app supports multiple languages, audio narration, and interactive quizzes to boost engagement and understanding. It features AR visualization, offline functionality, user-friendly interface, and marker-based tracking for 3D models. Tested in classroom-like settings, results showed improved student motivation and comprehension compared to traditional 2D methods. Importantly, its offline capability and language support make it especially valuable for rural or low-connectivity areas, ensuring equal access to quality, personalized, and culturally relevant education. Dhanya and Ghamalar (2025). Sipahutar and Handayani (2022) developed an offline English learning app for junior high school students to address difficulties in learning English as an

international language. Built with Adobe Animate CC 2017 and based on the curriculum, the app was designed using UML modeling and tested on smartphones. Results show it makes learning easier, engaging, and functional, helping teachers deliver material effectively while ensuring access without internet.

This study of Dhivya et.al (2022) introduces LearnHub, an offline-first e-learning mobile app designed to address connectivity challenges in education. The app allows students to download lessons, videos, and PDFs for offline use, while offering multiple courses, interactive quizzes, and gamification features like XP, badges, and levels to boost motivation. Built with Flutter for cross-platform compatibility and SQLite for local storage, LearnHub was compared with existing e-learning systems and found to enhance engagement and content accessibility. Results highlight its strength in inclusivity and offline learning, with future plans to add an offline AI assistant and an admin dashboard for dynamic content management.

Performance Monitoring

Educational Data Mining for Monitoring and Improving Academic Performance at University Levels, Educational Data Mining on 712 Moodle LMS logs from an African University to analyze student and staff usage patterns and their impact on academic performance. Data was processed using machine learning algorithms (clustering, classification, visualization) in WEKA, while SPSS correlation analysis examined links between course tool usage (Quiz, Assignment, Forum, Chat, Files, etc.) and final grades. Results showed significant correlations at the 0.01 level, indicating that higher engagement with LMS resources positively

influenced student performance. Findings highlight the strategic value of LMS data for academic planning and improving learning outcomes. Okike and Mogorosi (2020). Costa et.al (2020) developed Monitoring Academic Performance Based on Learning Analytics and Ontology: A Systematic Review

the coordinated use of Learning Analytics and Computational Ontologies in evaluating student performance. From 1,230 studies, 31 were analyzed focusing on their use in describing Taxonomies of Educational Objectives and applying analytical models to educational data. Results show few studies connect LMS interactions to learning experiences, most use ontologies for learning object assessment and sequencing, and none report coordinated use for performance monitoring. The review concludes that future research should develop new models integrating both approaches to improve academic performance evaluation.

The study of academic performance monitoring system using least squares regression in MATLAB to predict student outcomes. A GUI was designed to forecast relationships between factors, with accuracy measured through Percentage Forecast Error (PFE) and Mean Absolute Percent Error (MAPE). The sample included 182 BS Information Technology students, with 13 selected via random sampling and their GPAs used for prediction. Results show the system effectively monitors and forecasts performance, supporting improved learning outcomes and academic planning. Pilarca(2023). The study developed of a web-based centralized tracking system for academic performance to address issues in traditional methods such as fragmented data, manual errors, and lack of real-time access. The system provides a unified platform for storing and managing student

records, assessments, and progress reports, ensuring accessibility, scalability, and secure data handling. Findings highlight its role in enhancing transparency, identifying learning gaps, supporting outcome-based education, and generating analytical reports. Overall, web-based tracking systems modernize academic management and contribute to continuous quality improvement in higher education. Karper et.al (2025).

The limitations of Learning Management Systems (LMSs) in monitoring student progress and proposes the use of Learning Analytics with Student Performance Indicators (SPIs). Conducted at Cadi Ayyad University in Morocco, the case study tracked students in a Web Technologies course using SPIs to analyze instructional progress. Findings show that SPIs help faculty identify underperforming students, anticipate those at risk, and provide timely interventions. The research highlights that transforming LMS data into meaningful indicators enhances learner self-regulation, engagement, and academic support. Qazdar et.al (2023).

Internet Connectivity in Rural Education

A qualitative study of Pradana and Josiah (2024) in rural schools in Indonesia found major barriers to technology use, including poor internet, limited devices, and unstable electricity. Teacher readiness was also critical, with many lacking digital skills. Despite these challenges, successful integration of technology improved student engagement and outcomes. The study highlights the need for stronger infrastructure and teacher training to reduce rural–urban disparities and maximize educational benefits. A ICT integration in rural higher education

highlights persistent challenges such as poor internet access, outdated infrastructure, high costs, and low digital literacy among students and faculty. These barriers limit equitable learning opportunities and curriculum relevance, while rapid technology obsolescence worsens disparities. However, solutions like infrastructure upgrades, financial support, digital literacy programs, customized curricula, and sustainable technology adoption can help rural institutions harness ICT effectively. Kumari and Srivastaba (2023).

Donald and Hashim (2025) indicated the rapid advancement of technology has revolutionized the education sector, presenting new opportunities for teaching and learning. In rural Malaysia, especially Sabah, schools face inadequate infrastructure, poor internet, limited digital tools, and lack of teacher training, which widen the digital divide. A qualitative study revealed teachers' perspectives on these barriers, noting that while some successful implementations improved engagement, addressing infrastructure gaps and providing professional development are essential for equitable digital learning. The study of Raw and Baji (2020) people in rural India face greater scarcity of facilities compared to urban areas, limiting their ability to benefit from technological advancements. ICT offers potential solutions through satellite, web, and mobile applications, especially in education, which can help bridge the urban–rural divide. However, case study findings show that inadequate infrastructure, inefficiencies, and systemic issues hinder smooth ICT implementation, causing delays and high costs that make solutions infeasible. The study stresses the urgent need for strong collaboration

between public and private partners to improve infrastructure and ensure effective ICT roll-outs in rural education.

The strategies for closing the digital gap in rural education by exploring cost-effective internet connectivity models such as community Wi-Fi, satellite broadband, and mobile education services. Using a mixed-methods approach, the research analyzes connectivity models alongside educational achievement data, showing that infrastructure development and community participation enhance learning outcomes. The findings emphasize the importance of comprehensive policies, collaborative initiatives, and private investments to achieve sustainable results. Overall, the study supports global efforts to promote equal educational opportunities through technology in underserved areas. Mitra and Shah (2024).

Synthesis of Related Reviewed Literature and Studies

The reviewed literature highlights the growing integration of Artificial Intelligence (AI), Learning Management Systems (LMS), offline educational applications, and performance monitoring systems in improving modern education. These studies collectively emphasize the importance of technology in enhancing accessibility, personalization, and effectiveness of learning environments.

Several studies discuss the role of Artificial Intelligence in knowledge management and educational systems. Studies show that Artificial Intelligence enhances knowledge management and educational systems by improving knowledge sharing, communication, and organizational performance (Pai et al., 2021). AI-based tools like Smart Assist provide personalized learning through NLP, machine learning, secure access, and 24/7 support (Swetha et al., 2025).

Frameworks such as LAMB enable the integration of AI assistants into LMS platforms with privacy and reliability (Forment et al., 2025). Additionally, AI-powered LMS platforms support coding education through chatbots, project guides, and career resources, improving student skills and employability (Raghavendrachar et al., 2023).

Research highlights that Learning Management Systems enable flexible, accessible, and organized digital learning environments (Turnbull et al., 2020). LMS platforms help manage learning materials, track student progress, and improve communication (Rosario and Dias, 2022). They also enhance usability, satisfaction, and learning quality (Rabiman et al., 2020). However, LMS usage does not always significantly impact academic performance, indicating the need for further improvement (Tus et al., 2021).

Another significant theme in the literature is the development of offline learning applications to address connectivity challenges. Studies emphasize that offline learning applications address connectivity issues by allowing students to download and access materials without internet access (Laksmi et al., 2025). These systems support self-paced learning, improve accessibility, and increase engagement (Ahillon Jr. and Aquino, 2024). Applications like EduVision AR enhance motivation through interactive features (Dhanya and Ghnamalar, 2025), while LearnHub uses gamification and offline access to improve engagement and inclusivity (Dhivya et al., 2022).

Literature shows that academic performance monitoring systems use data analytics and predictive models to track student progress and improve outcomes

(Okike and Mogorosi, 2020). Learning analytics and ontologies support better evaluation of learning processes (Costa et al., 2020). Systems using regression models help predict performance and support planning (Pilarca, 2023), while centralized tracking systems improve transparency and data management (Karper et al., 2025). Student Performance Indicators also help identify at-risk students and enable early intervention (Qazdar et al., 2023).

Studies reveal that rural education faces challenges such as poor internet access, limited devices, unstable electricity, and low digital literacy (Pradana and Josiah, 2024). Infrastructure limitations and high technology costs further hinder ICT integration (Kumari and Srivastaba, 2023). Lack of teacher training also contributes to the digital divide (Donald and Hashim, 2025), while inefficient ICT implementation limits benefits (Raw and Baji, 2020). However, solutions such as infrastructure development and cost-effective connectivity models like community Wi-Fi and satellite broadband can help improve access (Mitra and Shah, 2024).

NOTES

Learning Management System

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CHAPTER III

METHODOLOGY

In this chapter it presented the methods and procedure employed by the researchers to conduct the study, aligning with the project's objectives. It outlined and elaborated on the specific actions and tasks undertaken to complete the project.

Research Design

The researcher uses a development and descriptive method approach in obtaining the system desired in this study. The researcher utilized a range of data collection techniques such as interviews using Direct/Interview method, survey, observations, and analysis of existing documents. The methods used by the researchers enabled to gather and acquire the essentials information and data necessary of constructing the study and understanding the questions.

Research Methodology

The Agile Model is a process model within the Software Development Life Cycle (SDLC) that emphasizes iterative development, customer collaboration, and flexible responses to change. Liyanage (2021). Agile software development goes beyond specific frameworks like Scrum, Feature-Driven Development (FDD), or Extreme Programming (XP), and beyond practices such as Pair Programming (PP), Test-Driven Development (TDD), stand-ups, sprints, and planning sessions. It serves as an umbrella concept that encompasses a variety of methods and techniques, all rooted in the values and principles outlined in the Agile Manifesto. The essence of Agile lies in applying these principles to guide decisions based on

the context at hand. What sets Agile apart from other approaches is its emphasis on collaboration solutions emerge through teamwork among self-organizing, cross-functional groups that select and adapt practices suited to their situation. Trivedi (2021). By applying the Agile programming method, the researchers were able to simplify and improve the process of developing both the application and the web-based system. This approach encouraged transparency, upheld quality standards, and ensured that the project stayed organized and detailed. Agile also allowed for faster development cycles, better adaptability to changing requirements, stronger teamwork, and a higher-quality final product through continuous feedback and step-by-step improvements.

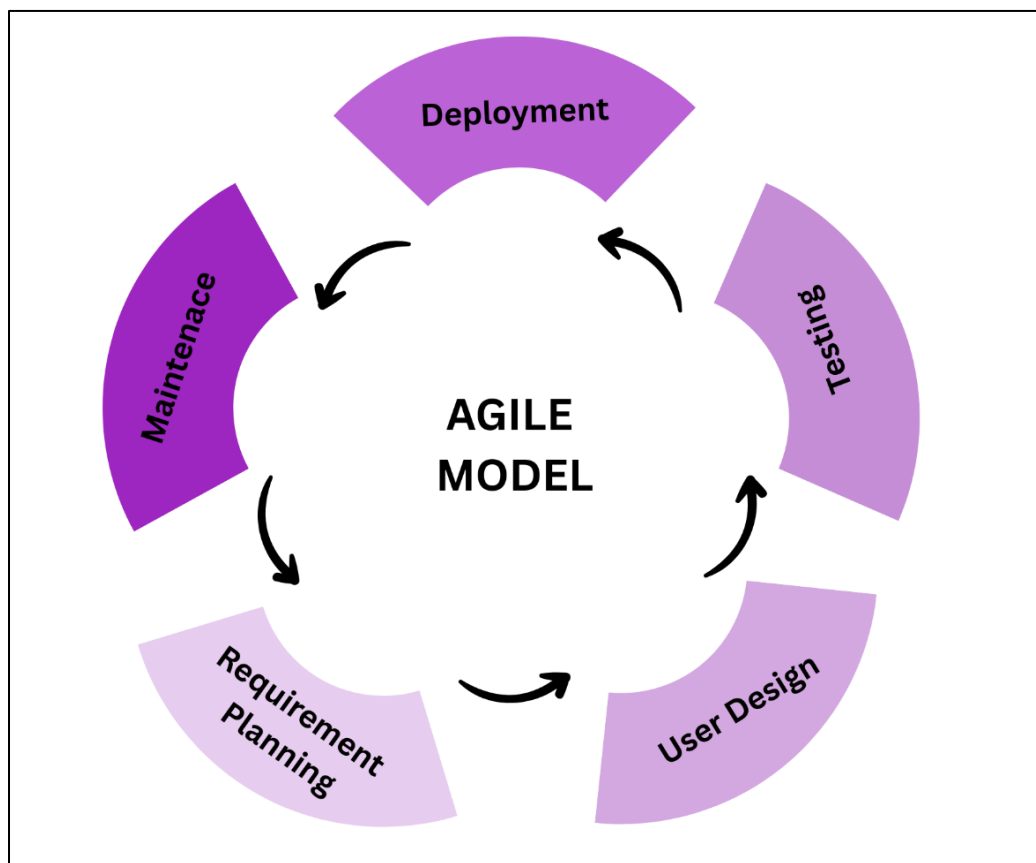


Figure 3.1
Agile Software Development Life Cycle Model

Phase 1: Requirements Planning

During the initial phase of project planning, the researcher listed the key requirements and identified possible challenges. They also prepared proactive solutions to handle these challenges during the development process.

The requirements planning phase began with consultations at Palangon High School, where the researcher engaged with the school head, teachers, and the identified client. Through these discussions, the researcher identified the school's pressing challenges, particularly the lack of stable internet connectivity, reliance on printed modules, and fragmented methods of file distribution such as USB drives and messaging applications. These issues hindered effective teaching and learning, consumed significant time and resources, and prevented timely identification of struggling students.

Teachers emphasized the need for a centralized system that would allow them to upload modules, quizzes, and activity sheets while also monitoring student performance in one platform. Students expressed the importance of consistent access to learning materials and support mechanisms that could help them comprehend lessons more effectively. Administrators highlighted the necessity of a cost-efficient solution that reduces reliance on printed modules and promotes sustainability. These consultations established the foundation for the LearnSync project, ensuring that the system's design directly responds to the school's needs.

From these the functional requirements were defined teachers must be able to upload text-based materials, create quizzes, assign activities, and record scores, while students should access these resources and interact with an AI assistant that provides summaries, guided explanations, and adaptive feedback

strictly from teacher-uploaded content. Non-functional requirements included offline accessibility through LAN/Wi-Fi, scalability for low-resource environments, accuracy of AI responses, and ease of use for both teachers and students.

Figure 3.1 show as the hardware requirements to successfully implemented and utilized the system. Computer hardware refers to the physical components that make up a computer system and enable it to perform essential functions, such as input, output, processing and storage. Susnja and Smalley (2026).

Table 3.1
Hardware Requirements for **LearnSync: An Offline Lan-Based Learning Management System with AI Academic Assistant**
Hardware Requirements

Hardware	Minimum	Recommended
Processor	At least Core i3	Core i5 and above, latest generation
RAM	8GB	16 GB or more
Storage	128 GB SSD	256 GB SSD or higher

The processor, RAM, and storage each play a distinct role in supporting the system's performance. The processor acts as the central unit that enables the learning management platform and the AI academic assistant to run smoothly. While a Celeron processor can handle basic tasks such as uploading modules and accessing files, a Core i3 or higher is recommended to efficiently process AI assistant, generate summaries, and provide adaptive feedback without delays. RAM, on the other hand, ensures responsiveness and multitasking capabilities. With multiple students and teachers accessing the system simultaneously through the LAN, a minimum of 8 GB allows basic functionality, but 16 GB or more is

necessary to prevent lags and ensure real-time performance when handling quizzes, modules, and AI responses.

Storage capacity is equally critical, as it houses lesson modules, quizzes, student records, and AI training data. An SSD provides faster file access compared to traditional HDDs, ensuring quick retrieval of instructional materials. While 128 GB is sufficient for initial deployment, 256 GB or higher is recommended to accommodate scalability, supporting larger datasets, more subjects, and future expansion of the system. Altogether, these hardware requirements guarantee that LearnSync can function as a lightweight, offline AI-powered learning management system tailored to low-resource environments. Meeting the minimum specifications ensures usability, while the recommended specifications provide sustainability and scalability, aligning with the project's goal of bridging digital divides and enhancing teaching and learning.

Table 3.2
Software Requirements for **LearnSync: An Offline Lan-Based Learning Management System with AI Academic Assistant**
Software Requirements

Software	Minimum	Recommended
Operating System	Windows 10/Android 10	Windows 11/android 10 and above
Internet	None, Local Area Network	
Language	Python, Java, html, CSS, JS	
Database	mariaDB	

Table 3.2 Software requirements needed to ensure compatibility and run the system efficiently and effectively. software, instructions that tell a computer

what to do. Software comprises the entire set of programs, procedures, and routines associated with the operation of a computer system. Britannica (2026).

The system design to ensure compatibility and efficient operation, the system requires specific software components. At minimum, the operating system should be Windows 11 or Android 10 and above, with the latest stable release recommended for enhanced security and performance. The system can function without internet access by relying on a Local Area Network, though a stable LAN or broadband connection is advisable for smoother data transfer and real-time interactions. The programming languages used in development include Python, Java, HTML, CSS, and JavaScript, and it is recommended to utilize updated versions of these languages to maintain compatibility with modern frameworks and libraries. For data management, MariaDB serves as the database system, providing reliable storage and retrieval of information, with the latest stable release recommended to ensure optimal performance and security. Overall, these software requirements are essential to guarantee that the system runs effectively, supports scalability, and delivers a seamless user experience.

. The researcher developed a Gantt chart, presented as Figure 3.2, to serve as a guide for project implementation. This chart highlighted the key milestones necessary to ensure that tasks were completed within the designated timeframe and that the overall project advanced according to the deadlines set by the researcher.

Activities	Month	January				February				March				April			
	Week	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Brainstorming, identification of possible topics																	
Identification of Possible Client																	
Client Interview and Data Gathering																	
Analysis of Gathering Data																	
Proposing of Titles																	
Drafting of Chapter 1-3																	
Design And Development Phase																	
Planning of System Design																	
Analysis of Main Function																	
Designing of Database and Initial Interface																	
Proposal Defense																	
Coding of the System																	
System Evaluation of Capstone Adviser																	
Drafting of chapter 4-5																	
Final Checking																	
Pre-final Defense																	
Revision on Documents																	
Revision on Documentation																	
Final Defense																	
Implementation																	
Turnover of the System																	
Submission of Final Copies of Chapter 1-5																	

Figure 3.2
Gant Chart of the Development of “**Learnsync: An Offline Lan-based Learning Management System with AI Academic Assistant**”

Phase 2: User Design

The user design phase focuses on creating a practical and user-friendly interface by involving end-users in the design process. Researchers and developers collaborate to identify user needs, workflows, and preferences, ensuring the system is intuitive and efficient. This stage often includes prototypes, wireframes, and mock-ups that allow users to visualize the system and provide feedback before full development. By engaging users early, potential issues can be addressed, resulting in a final product that is accessible, reliable, and aligned with its intended tasks.

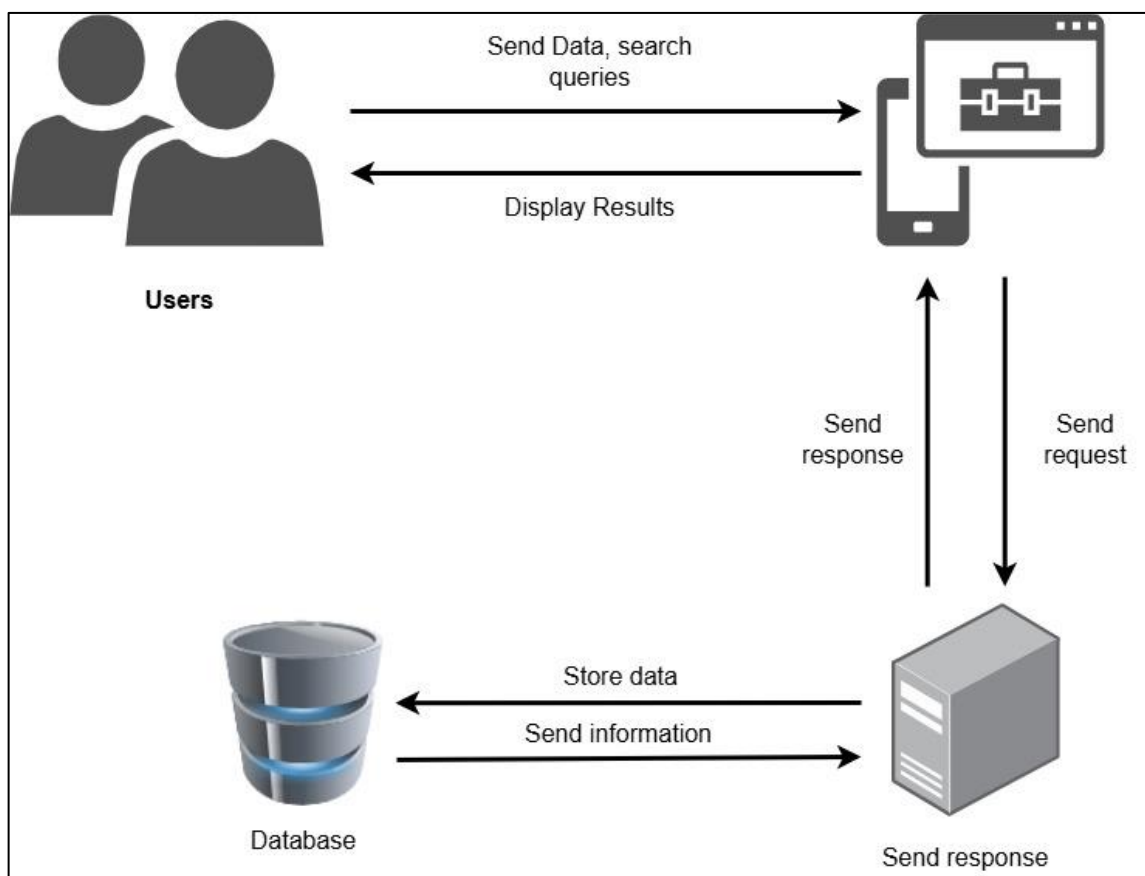


Figure 3.3

LEARNSYNC: AN OFFLINE LAN-BASED LEARNING MANAGEMENT SYSTEM WITH AI ACADEMIC ASSISTANT

Figure 3.3 illustrates the core architecture of the Learnsync an offline Lan-based learning management system with ai academic assistant. It highlights the main components involved in the interaction between the user and the system, including the devices, web system platform, server, database.

The process begins when the user (student or teacher) accesses the system through various devices such as smartphones, tablets, or desktop computers. These devices act as the interface, allowing students to interact with lessons, quizzes. Teachers use the same interface to upload content and monitor student progress. The client devices connect to the system via web browsers or LAN-enabled applications. They handle user inputs such as code submissions, lesson answers, or gameplay actions. These inputs are packaged into requests and sent to the server for processing. In LearnSync's offline model, this connection happens through the school's local Wi-Fi network, ensuring accessibility without internet dependency.

The server acts as the central processing unit of the system. It receives requests from client devices, processes the application logic checking quiz answers, running AI feedback, and manages communication between the user interface and the database. In LearnSync, the server also hosts the AI academic assistant, which provides adaptive feedback, guided explanations, and performance monitoring all while functioning offline. While the database stores all persistent information, including user profiles, lesson modules, code submissions, progress tracking, and quiz results. It communicates with the server to retrieve and update data in real time, ensuring accuracy and availability across the system. For

LearnSync, this centralized database eliminates fragmented storage methods like USB drives or messaging apps, making instructional materials organized, accessible, and reliable.

This guide shows how the system flows user's client devices server database back to users. Each component plays a vital role in ensuring smooth interaction, centralized resource management, and efficient learning delivery. This architecture is adapted to an offline LAN-based model, directly addressing Palangon High School's connectivity challenges while still providing modern AI-driven educational support.

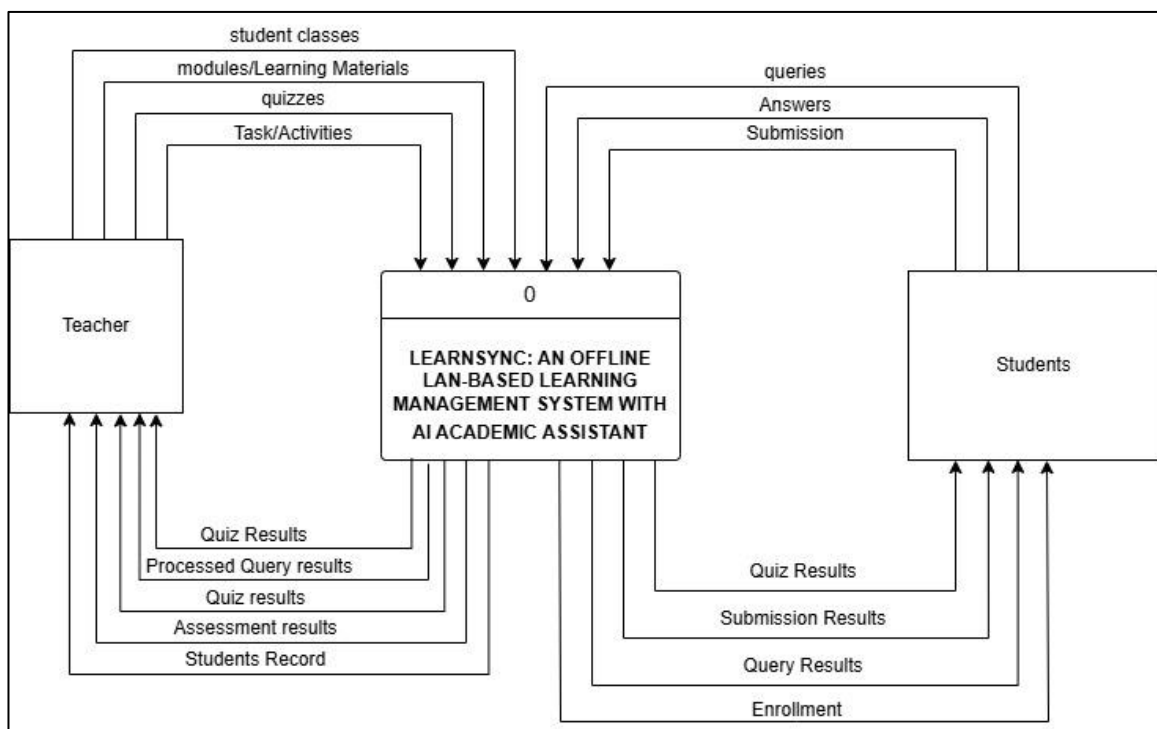


Figure 3.4

Context Diagram of Learnsync: An Offline LAN-based Learning Management System with AI Academic Assistant

The 3.4 context diagram of LearnSync: An Offline LAN-Based Learning Management System with AI Academic Assistant shows how teachers and

students interact with the system through a local network. At the center is the LearnSync platform, which acts as the hub for communication, data storage, and AI assistant support.

On the teacher's side, the system allows them to manage classes, upload modules and learning materials, create quizzes and activities, and record student results. Teachers also receive processed outputs such as quiz results, submissions, assessment records, and AI query responses, which help them evaluate performance and provide feedback.

On the student's side, learners connect to the system to access enrollment information, modules, and activities. They submit answers, upload tasks, and attempt quizzes. Students also benefit from the AI assistant by sending queries and receiving contextual responses that guide their learning. The system returns results such as quiz scores, submission feedback, and query outputs, ensuring that students can track their progress.

Overall, the diagram highlights a two-way flow of information: teachers provide content and assessments, while students engage with the materials and submit work. The LearnSync system processes these interactions, integrates AI support, and delivers results back to both sides. This design emphasizes efficiency, offline accessibility, and automated feedback loops, making it a practical solution for rural or resource-limited schools where internet connectivity may be unreliable.

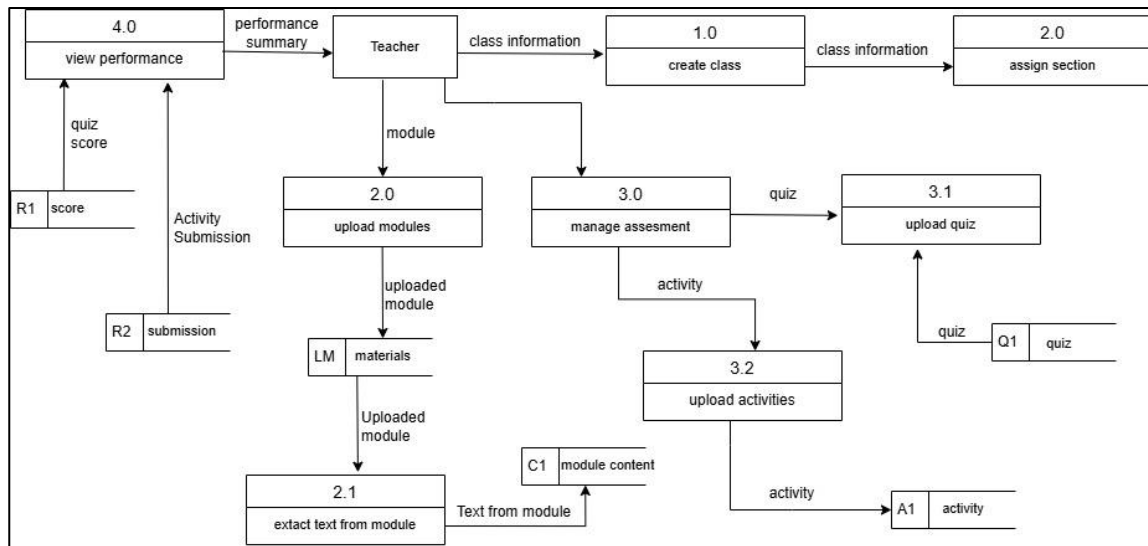


Figure 3.5
Data Flow Diagram for Teacher

Figure 3.5 illustrates the Data Flow Diagram (DFD) of the Teacher Module, showing how data moves within the system and how the teacher interacts with different processes and data elements. The diagram presents the core instructional and administrative functionalities of the module, beginning with class creation and section assignment, followed by the uploading of modules, management of assessments, and monitoring of student performance.

The process starts with Create Class (1.0), where the teacher establishes a new class to organize students and learning materials. Once the class is created, the teacher proceeds to Assign Section (2.0), grouping students into appropriate sections for better management. After sections are assigned, the teacher can Upload Modules (2.0), which serve as the primary learning materials (LM). Within this process, the system can also Extract Text from Module (2.1), converting module content (C1) into usable formats for for retrieval of data by AI Assistant.

The next major function is Manage Assessment (3.0), where the teacher oversees different evaluation tools. This includes Upload Module (3.1) for

exchanged among processes and data stores to support learning activities. The flow begins with the enrollment process, where the student initiates participation through Enroll Class (1.0) and submits details via Submit Enrollment (1.1). These records are stored in the Enrollment (A1) data store and linked to the Class (C1) data store, establishing the student's access to the course.

Once enrolled, the student may interact with the AI assistant through Query AI (2.0). The query context is retrieved from the enrollment data and processed by AI Generation (2.1), where the Language Model (LM) provides tailored responses to support learning. This step highlights the integration of AI into the system, enabling students to receive contextualized guidance and assistance that complements traditional instruction.

The next stage involves quiz interaction, where the student answers assessments through Answer Quiz (3.0). Responses are recorded in the Student Answer (Q1) data store and evaluated by Check Quiz (3.1), producing a Score (R1) that delivers immediate feedback. Beyond quizzes, students engage in activity submission and teacher evaluation. Through Submit Activity (4.0), assignments are forwarded to Teacher Evaluation (4.1), where teachers assess the work. Results are stored in the Submission (R2) data store, and students can access both their scores and submissions via View Score (4.1) and View Submission (4.2), ensuring transparency in evaluation.

The system supports performance monitoring through View Performance (5.0), which consolidates scores, submissions, and AI interactions into a comprehensive overview of student progress. Overall, the DFD demonstrates a

structured and integrated flow of data between the student, AI system, and teacher. It highlights how the platform supports interactive learning through enrollment management, AI-assisted queries, automated quiz scoring, teacher-led evaluation, and performance tracking. This ensures a balanced environment where automation enhances efficiency while teachers provide qualitative oversight, resulting in a comprehensive and student-centered educational experience.

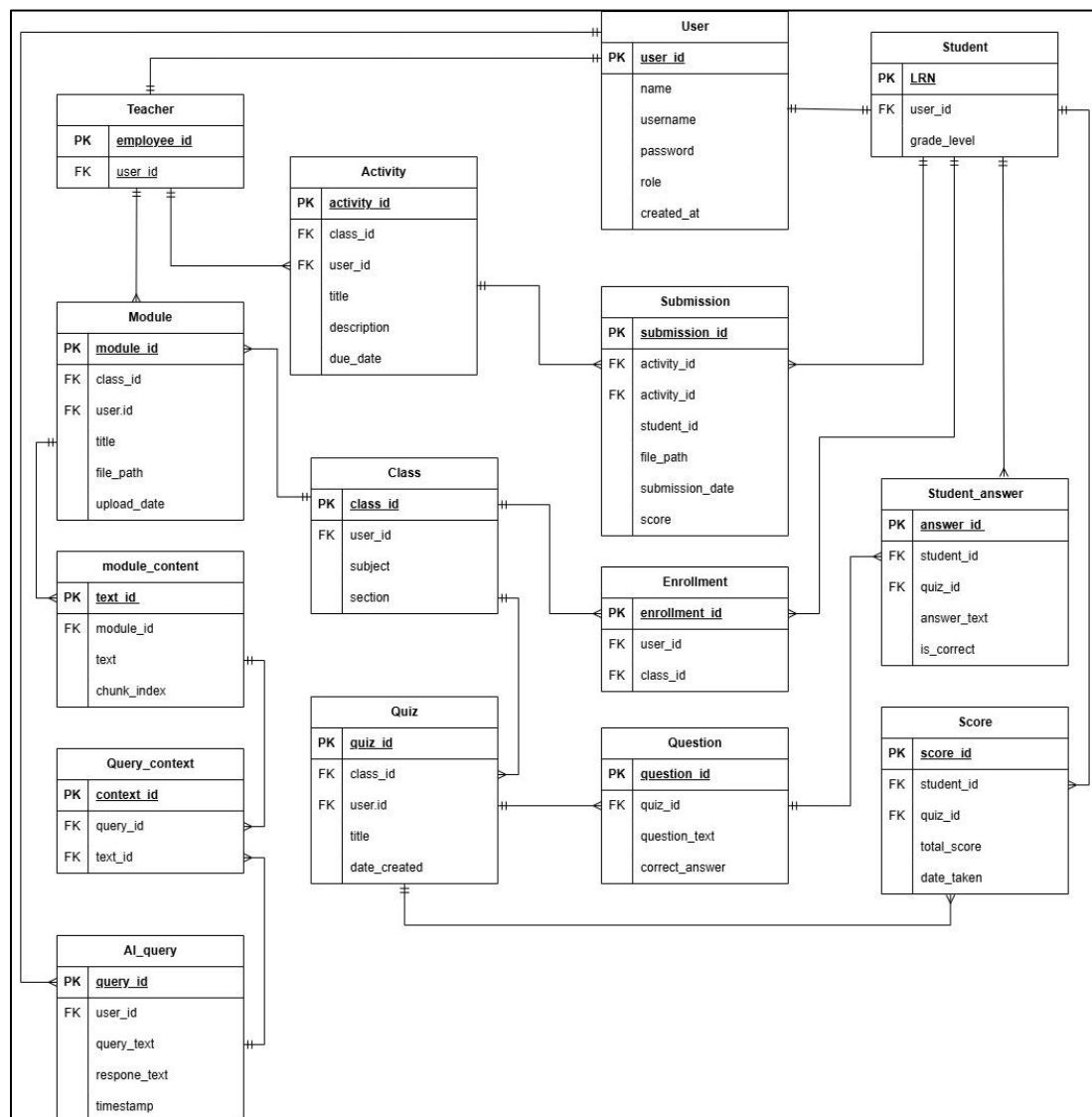


Figure 3.7
Entity Relationship Diagram of “Learnsync: An Offline Lan-based Learning Management System with AI Academic Assistant”

The Entity-Relationship Diagram (ERD) of the Learnsync: An Offline Lan-based Learning Management System with AI Academic Assistant illustrates how the database is organized to support teaching, learning, and assessment. At the center of the design is the Users entity, which stores essential information such as login credentials, names, roles, and account creation dates. This central table connects directly to both Teachers and Students, ensuring that every participant in the system is properly identified and managed. Teachers are linked to classes they handle, while students are tied to their accounts through their Learner Reference Number (LRN).

The academic structure is represented by the Classes and Enrollment entities. A class is defined by subject and section, and it is associated with a teacher. Students join classes through the enrollment table, which serves as a bridge between learners and their assigned subjects. Learning materials are organized in the Modules entity, which stores uploaded files and metadata such as titles and dates. To enhance accessibility, module content is broken down into smaller chunks through the Module content entity, allowing for easier retrieval and processing.

Learning activities are managed through the Activities entity, which contains assignments and tasks created by teachers. Student responses are stored in the Submissions entity, including file paths, submission dates, and scores. Assessments are handled by the Quizzes entity, which is linked to specific classes and teachers. Each quiz contains multiple Questions, and student responses are recorded in the student answers entity, marking whether answers are correct. Final

results are summarized in the Scores entity, which tracks total scores and completion dates.

Beyond traditional learning, the ERD also integrates AI support through the AI query and Query context entities. These tables log user queries to the AI assistant, store responses, and connect queries to relevant module content, enabling intelligent, context-aware assistance.

The ERD demonstrates a well-structured system where users are at the core, classes and enrollments define academic organization, modules and activities deliver content, quizzes and scores measure performance, and AI queries provide additional support. The design ensures data integrity, scalability, and complete tracking of student progress, making it an effective backbone for a modern educational platform.

NOTES

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